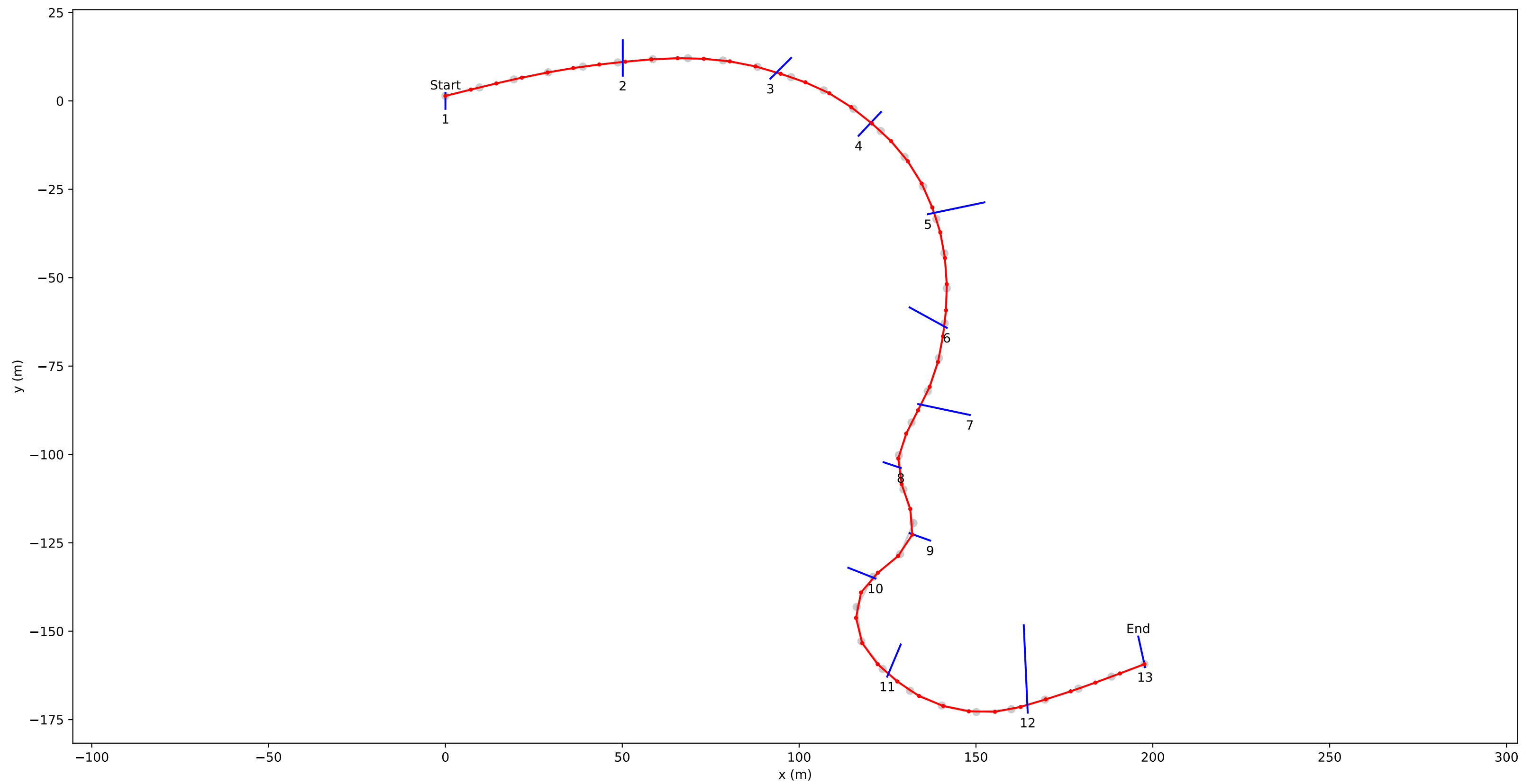


### Map plot: optimized path through gates



## Track Generation Report

=====

Generated: 2026-05-30 21:45:09

Gate file: ref\_gates\_2\_gates.csv

Gate count: 13

Saved DOFs file: opt\_dofs\_2026-05-30\_21-45-09.npy

### Optimization parameters

method: L-BFGS-B

maxiter: 500

ftol: 1e-07

### Point counts

objective function points: 52

path generation points: 39

### Results

total optimization time: 494.735 s

total path length: 377.6906 m

**total objective function value: 0.309277**

Per-step time estimate (one measured objective evaluation)

make spline from dofs: 3.90 ms

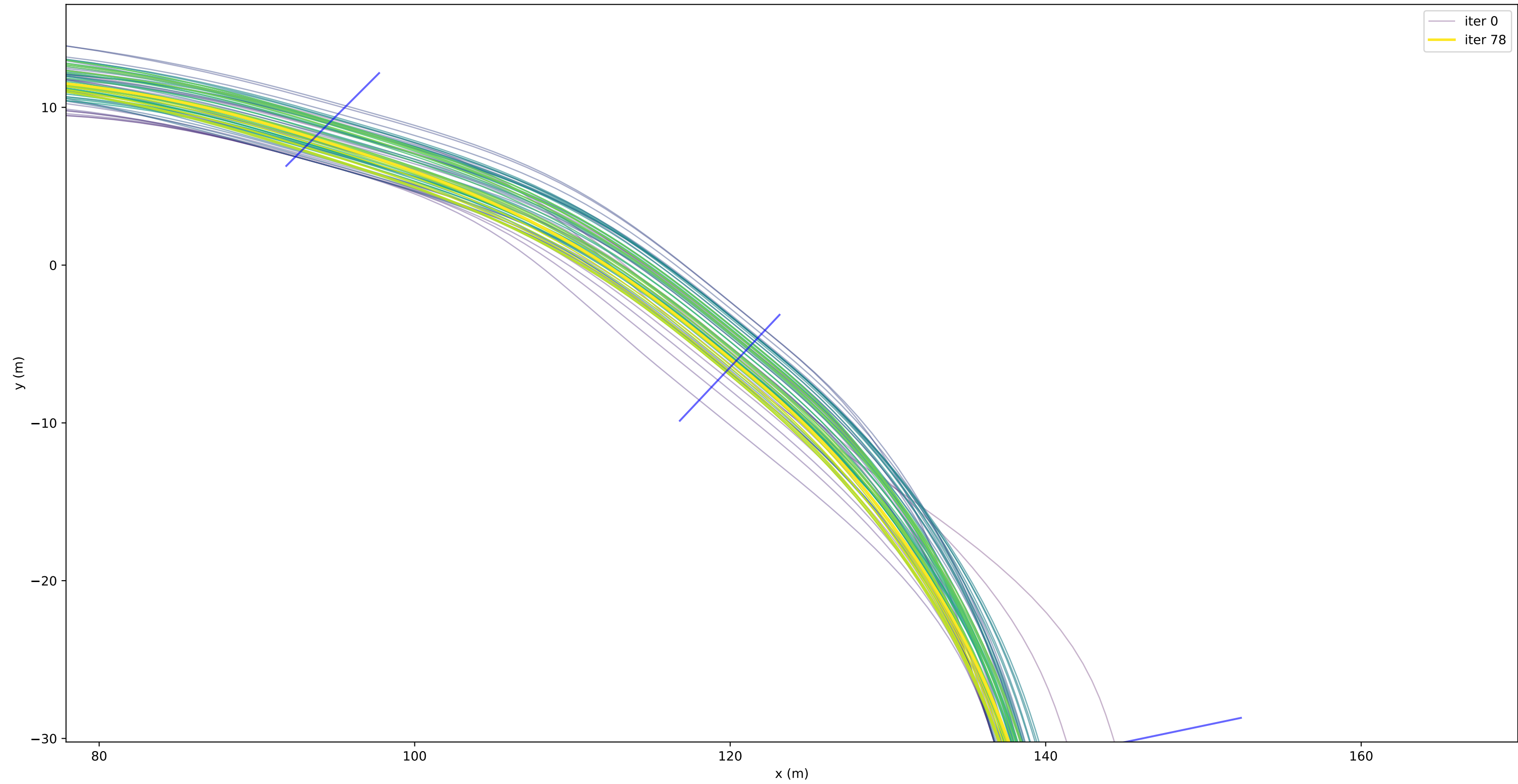
path length function: 0.84 ms

equidistant times generated: 0.51 ms

curvature objective function: 38.96 ms

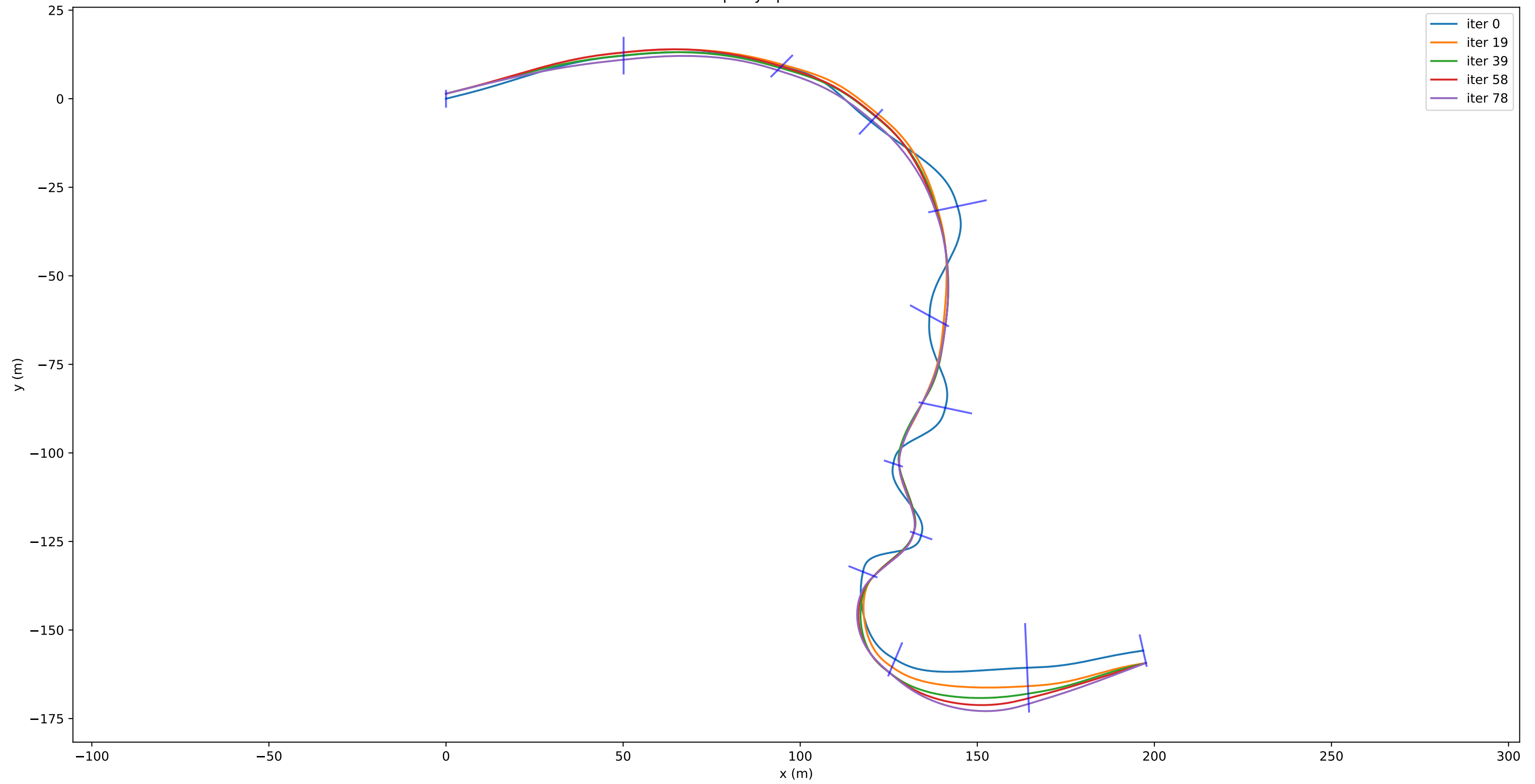
Optimization progression

Optimization progression (79 iterations)



Equally spaced iterations

5 equally spaced iterations



Instant radius vs distance

